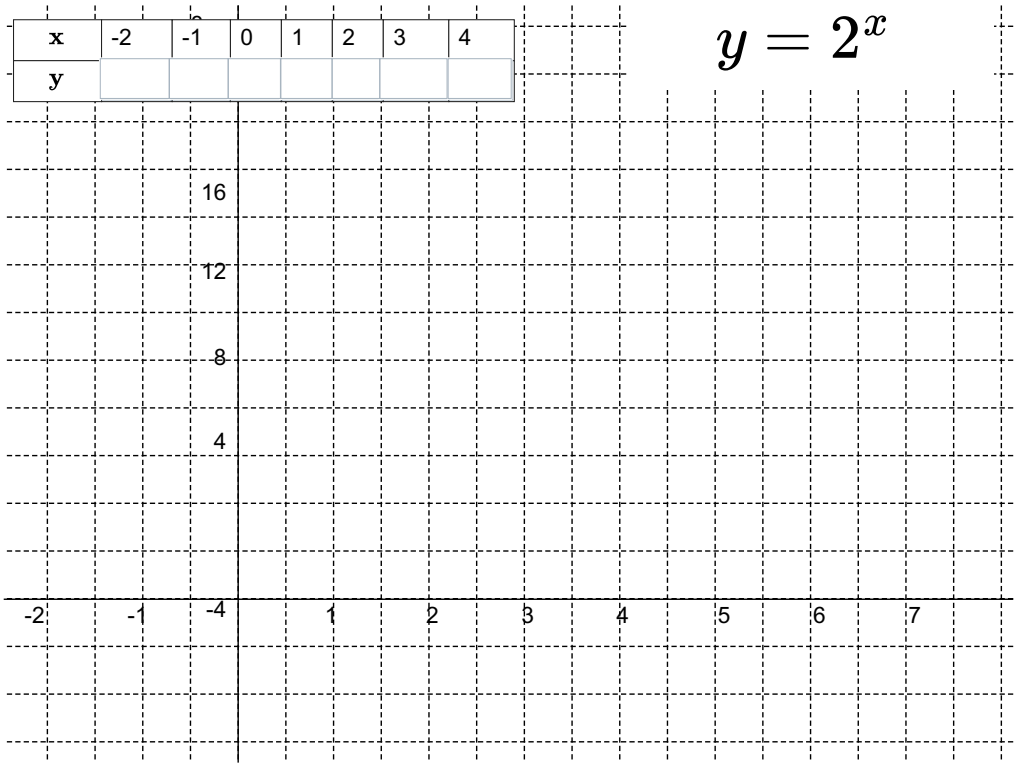
1.24 Graph Sketching: Exponentials and Logarithms

Chapter 14 · Original slide 4

x	-2	-1	0	1	2	3	4
y							

$y = 2^x$



Chapter 14 · Original slide 5

Contrasting exponential graphs

5

On the same axes sketch $y = 3^x, y = 2^x, y = 1.5^x$

On the same axes sketch $y = 2^x$ and $y = \left(\frac{1}{2}\right)^x$

1.24 Graph Sketching: Exponentials and Logarithms

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Graph Transformations

6

Sketch $y = 2^{x+3}$

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Exercise 14A (Pages 313-314)

7

6 The graph of $y = ka^x$ passes through the points $(1, 6)$ and $(4, 48)$. Find the values of the constants k and a.

7 The graph of $y = pq^x$ passes through the points $(-3, 150)$ and $(2, 0.048)$.

a By drawing a sketch or otherwise, explain why $0 < q < 1$.

b Find the values of the constants p and q.

1.24 Graph Sketching: Exponentials and Logarithms

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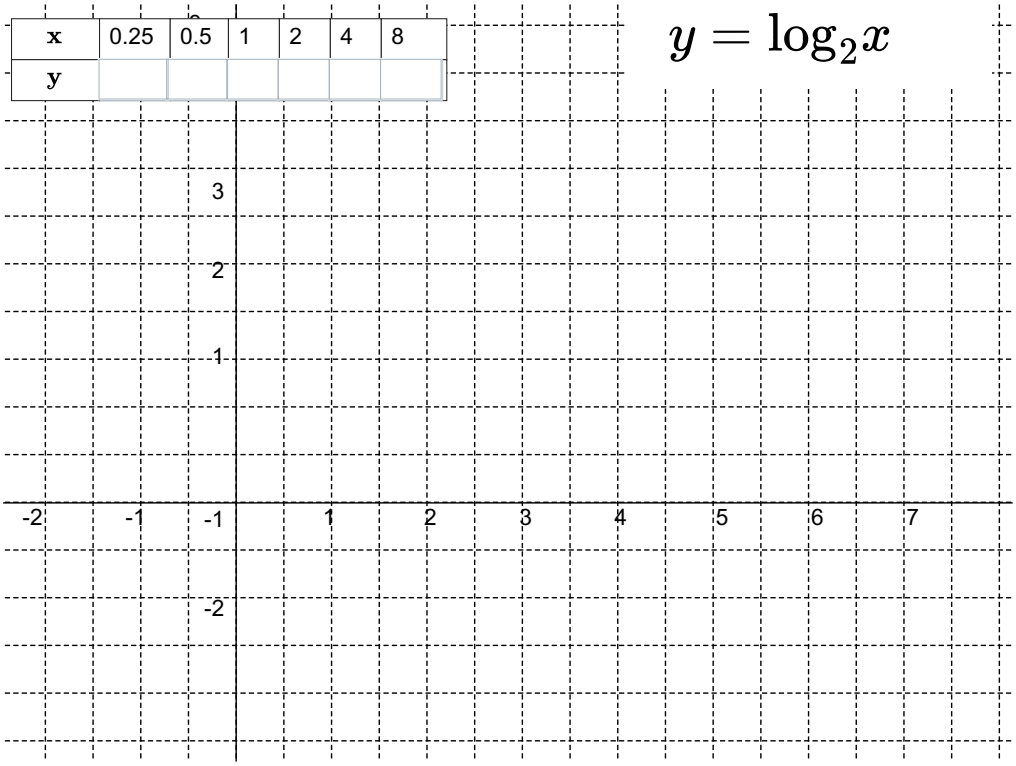
x	0.25	0.5	1	2	4	8
y						

$y = \log_2 x$